

Crowd-out and Competition: How non-governmental development programs shape public spending

Blaine Finstein

Abstract

Throughout the developing world, aid programs provide basic goods, but these are offered alongside public services and can reasonably be attributed by citizens to the state. I argue that external development programs substitute for public spending, providing discretionary resources that incumbents can award strategically. However, political competition disciplines the capacity for politicians to exploit them. Leveraging a randomized evaluation of a World Bank-financed community-driven development program in the Philippines, I show that the program reduced public spending in the same domains and that politicians reallocated fiscal slack to political supporters. However, this effect is concentrated in low-competition municipalities where incumbents were insulated from electoral pressure. Suggestive evidence indicates that public resource reallocation may have attenuated welfare returns from the program. These results demonstrate that aid can crowd out public spending even when resources bypass government budgets and clarify a long-standing empirical puzzle in fungibility literature. Aid shapes public spending, but democratic incentives determine whether this complements or corrodes the state.

1 Introduction

Foreign aid constitutes one of the most significant external forces shaping governance and welfare in the developing world. Each year, international donors and non-governmental organizations (NGOs) finance billions of dollars in projects designed to reduce poverty and improve human welfare. These interventions can yield substantial welfare gains. Yet, they also alter the incentives of governments that operate in the same service domains. When external actors assume responsibility for providing public goods, politicians may reduce spending and redirect resources elsewhere (Bräutigam and Knack (2004); Moss, Pettersson Glander, and Walle (2006)). As a result, programs that appear successful may inadvertently blunt their own long-term impact or weaken state capacity.

A large literature has extensively documented that intergovernmental grants and official development assistance can crowd out public spending (Feyzioglu, Swaroop, and Zhu (1998); Chatterjee, Giuliano, and Kaya (2012)). However, this research has overwhelmingly focused on funds delivered *to* governments. In practice, external aid programs frequently overlap with government mandates, offering services that public institutions might otherwise provide. Budget constraints relax regardless of whether funds actually flow through government channels. If politicians exploit this fiscal slack to reduce programmatic service provision, even well-designed aid programs may facilitate political capture and yield attenuated welfare gains.

Distinctively, non-governmental aid delivers fiscal slack under conditions of observational equivalence. Citizens cannot necessarily infer that the goods they receive through aid programs are not provided by their government and have limited basis for sanctioning incumbents who take advantage of the fiscal space. This weakens the electoral constraint that normally disciplines the behavior of politicians. Furthermore, when aid is given to governments, social programs are still delivered by elected officials even if the funds crowd out public spending. If external programs crowd out public spending, then government services are replaced by aid programs operated by unelected providers. Non-governmental aid therefore has important normative implications for democratic governance.

Aid is often considered insulated from political distortion when delivered outside of government budgets, but can it still reshape public institutions? How do non-governmental programs alter the incentives of elected officials, and under what conditions? In this paper, I trace the causal chain of external development assistance on public spending. I show that non-governmental development programs afford fiscal slack that politicians distribute politically. I demonstrate that these outcomes are conditioned on electoral competition, which can mitigate the unintended political consequences of external development assistance. Lastly, I offer suggestive evidence that this ultimately mutes the benefits of aid. These results bridge salient but previously distinct strands of the literature — fungibility, accountability, and distribution — squaring studies of foreign aid with divergent empirical findings. Aid crowds out public spending in some con-

texts but not others, and sometimes weakens the institutions to which it is delivered (Feyzioglu, Swaroop, and Zhu (1998); Hines and Thaler (1995); Moss, Pettersson Gelande, and Walle (2006)). I show that aid can erode recipient institutions through fungibility and political reallocation, but that this response depends crucially on local democratic incentives.

I exploit a randomized evaluation of an aid program to achieve causal leverage rare in these domains. I study these dynamics in the context of Kalahi-CIDSS (Kalahi), a World Bank-financed community-driven development program in the Philippines. Kalahi was randomly assigned across municipalities and disbursed block grants directly to citizen committees for small-scale development projects. I merge impact evaluation data with rich administrative data on quarterly municipal public spending and procurement, enabling causal inference on governance outcomes.

I advance three central findings. First, I show that external development assistance can crowd out local public spending. Kalahi significantly reduced local government expenditures in overlapping service sectors with no compensating increases in other spending categories. This finding broadens the scope of aid fungibility theory: crowd-out can occur even when financing bypasses government budgets.

Second, I demonstrate that electoral competition, the core institutional mechanism of democratic accountability, conditions crowd-out. Using municipal election returns, I show that crowd-out effects are concentrated in low-competition settings, while competitive municipalities maintain spending. This mechanism helps reconcile a long-standing empirical puzzle.

Third, I show that crowd-out is not passive fiscal contraction but active reallocation toward political ends. Using municipal procurement records linked to village-level electoral returns, I demonstrate that treated mayors directed public contracts toward their most supportive villages, while their counterparts in municipalities that did not receive Kalahi allocated procurement more programmatically. Individual-level data from the Philippine Labor Force Survey corroborate this pattern: treated municipalities exhibit significantly higher rates of casual government employment, consistent with patronage payroll expansion (Colonnelli, Prem, and Teso (2020)). Crucially, the same competition gradient that disciplines crowd-out also disciplines how freed resources are spent. Together, these results show that Kalahi did not merely reduce public provision, but reshaped resource distribution and diminished programmatic governance where accountability was already weakest.

Finally, I offer suggestive evidence on the welfare consequences of crowd-out. Exploiting pre-treatment budgetary constraints that limit the capacity for mayors to reallocate resources, I examine how fiscal discretion affected Kalahi's development outcomes. Endline survey data from the impact evaluation show that Kalahi's welfare gains were significantly attenuated by the capacity for mayors to reallocate public resources. This pattern suggests that non-governmental de-

velopment assistance reverberates through public institutions in ways that blunt its impact on beneficiaries.

Together, this causal chain is evidence of a political equilibrium in which non-governmental aid restructures rent extraction conditional on accountability. In electorally competitive settings, aid complements public provision. In less competitive ones, it frees resources for political reallocation. The effectiveness of non-governmental development programs thus depends on political institutions as well as program design. Where accountability is strong, external assistance can reinforce governance. Where it is weak, it can erode it.

This reveals a broader class of political questions about external service provision and democratic institutions. When non-governmental provision blurs responsibility for social services and offloads control from elected officials, does democratic accountability weaken? How does external aid interact with local political geography and inequality? Under what institutional conditions beyond competition does external provision distort public institutions? Such questions extend the study of aid effectiveness to political consequences and help identify conditions under which external provision may weaken democratic governance.

The paper proceeds as follows. Section 2 develops a theoretical framework linking non-governmental development assistance, fiscal substitution, political reallocation, and electoral competition. Section 3 describes the Kalahi program and details the empirical strategy. Section 4 reports the main results on municipal spending. Section 5 traces the destination of freed resources through procurement records and labor market data. Section 6 links these fiscal dynamics to welfare outcomes using household survey data from the impact evaluation. Section 7 concludes.

2 Theory

Throughout the developing world, public institutions and aid agencies offer similar services to a limited population of overlapping beneficiaries. Villages receive water purification or electrification projects installed by an NGO while those in other parts of the country enjoy clean drinking water and electricity provided by a government utilities system. Children are inoculated in UNICEF vaccination drives while enrolled in a public health insurance scheme. Households benefit from cash transfers, literacy campaigns, or health interventions financed by international donors, while governments fund similar initiatives through local budgets. These parallel structures of service delivery create a shared domain of welfare provision between aid and the state.

Non-governmental development assistance can thus alter the incentives of politicians. Aid offers an opportunity to reallocate limited fiscal resources elsewhere, and incumbents may rationally reduce public investment in sectors served by aid agencies. The result is crowd-out: aid replaces rather than complements government spending.

Furthermore, aid-financed goods are observationally similar to public provision. Citizens who receive an externally financed intervention have limited basis for inferring that their mayor’s own spending declined. This informational friction weakens the electoral sanction that would otherwise deter substitution. Attribution failure makes retrenchment electorally survivable. Voters observe that a well was built or a road was constructed, but not whether the project was executed by the local government or an external donor. Welfare is visible, but the institution responsible for it is opaque. This reduces the electoral cost of substituting external goods for public provision.

Non-governmental development programs bear important distributional consequences as well. By decoupling service provision from public spending, aid provides discretionary resources, and by decoupling retrenchment from electoral sanction, it affords the cover necessary to exploit them. Politicians can leverage these resources to consolidate power and insulate themselves from electoral accountability. They may do so through clientelism (Hicken (2011); Stokes (2009); Keefer (2007)), distribution to co-ethnics or co-partisans (Franck and Rainer (2012); De Luca et al. (2018); Solé-Ollé and Sorribas-Navarro (2008)), transfers to local elites or brokers (Stokes, Dunning, and Nazareno (2013)), or rent extraction (Shleifer and Vishny (1993); Olken and Pande (2012)).

However, political competition constrains the capacity for politicians to exploit these perverse incentives. Competition encourages programmatic resource allocation because incumbents must satisfy a broad coalition of voters (Lindbeck and Weibull (1987); Lizzeri and Persico (2001)). While narrow allocation is sufficient and attractive under low competition, electoral risk pushes incumbents toward programmatic and non-excludable provision (Harding and Stasavage (2014); Burgess et al. (2015); Ferraz and Finan (2011)). Furthermore, competition may also enhance the proper attribution of public goods. Challengers have an incentive to publicize the underperformance of incumbents and contest attempts to claim credit for external programs. By raising the electoral cost of crowd-out, competition determines the threshold at which political reallocation is compatible with electoral survival.

Beyond political incentives, reallocation requires capacity. Even an unaccountable mayor cannot redirect fiscal resources they do not control. Municipalities with greater fiscal discretion face fewer binding constraints on how freed resources are deployed. Practically, this includes larger own-source revenue buffers, lower debt burdens, and more flexible payroll arrangements. Discretion is thus a prerequisite for the political reallocation of freed funds. The welfare implications follow directly. Crowd-out should attenuate program returns most sharply where politicians have both the incentive to substitute and the fiscal room to act on it.

This framework reconciles seemingly contradictory findings. Economic models of substitution predict that aid will crowd out public spending, yet empirical studies frequently identify cases of fiscal complementarity. The theory proposed here suggests that both outcomes are possible and depend on democratic insti-

tutions.

This logic yields four testable implications. First, external development assistance reduces public service provision in the same domains. Second, external assistance reshapes public resource distribution toward targeted benefits rather than programmatic goods. Third, electoral competition mitigates both of these outcomes. Fourth, where mayors have greater ex ante fiscal discretion, crowd-out attenuates the welfare returns that the program would otherwise generate. The empirical analysis that follows tests these claims using data from a randomized evaluation of Kalahi, a World Bank-financed community-driven development program in the Philippines, examining how local governments restructure public resources across electoral environments in response to aid.

3 Data and Empirical Strategy

Established in 2003, Kalahi aims to reduce poverty and increase local participatory governance in the Philippines. Kalahi delivers grants to qualifying municipalities for small-scale community development projects, such as farm-to-market roads, health stations, and irrigation and drainage systems.¹ Communities determine how to allocate funds democratically in order to meet their particular needs. In addition to financing local development projects, Kalahi trains community members in choosing, designing, and implementing projects.

Unlike intergovernmental transfer programs in which funds flow through municipal accounts, Kalahi delivered block grants directly to community committees at the village level. Municipal governments were neither financial intermediaries nor implementing agents. Facilitators from the Department of Social Welfare and Development (DSWD) supported communities in project selection and design, but disbursement bypassed the municipal budget entirely. This structure is common to community-driven development programs globally.

From 2012-2015, Innovations for Poverty Action (IPA) conducted an impact evaluation of Kalahi aiming to evaluate the effects of the program on key socioeconomic and governance outcomes via a randomized controlled trial (RCT). The impact evaluation spanned 198 municipalities across all three of the main island groups of the Philippines (Luzon, the Visayas, and Mindanao). Treatment was assigned via a series of public lotteries conducted within provinces, with municipalities matched into pairs immediately prior to each lottery on four pre-treatment characteristics: poverty incidence, population, land area, and number of barangays. Within each matched pair, one municipality was randomly assigned to treatment and one to control, yielding 99 treatment-control pairs

¹Municipal poverty rate is the predominant qualification for Kalahi. In the Philippines, municipalities are rated on a class scale of 1 (wealthiest) to 6 (poorest). In order to receive Kalahi, 4th – 6th class municipalities must have a poverty incidence above the national average of 26.5%, based on the 2009 Small Area Estimates (SAE) of the National Statistical Coordination Board (NSCB). 1st – 3rd class municipalities qualify if they have a poverty incidence of 40% or higher (DSWD (2024)).

across 26 provinces (Beatty et al. (2018)). All specifications exploit this structure directly, using matched pair fixed effects to absorb time-invariant within-pair heterogeneity consistent with the randomization design. While the exact timing of grant allocation varied, municipalities selected for treatment received one grant per year from 2012-2015. These were calculated per municipality on the basis of its number of barangays, the smallest administrative unit in the Philippines and the level at which citizen committees were organized. While all barangays in a participating municipality were eligible to receive Kalahi, funds were allotted in accordance with project priorities.² Figure 1 displays a map of the location of treated and control municipalities across the Philippines.

Kalahi was found to substantially improve welfare outcomes. Infrastructure projects reduced the time to key services and the cost to get farm products to market, water-related projects reduced the time and cost to obtain it, and education initiatives increased school enrollments (Beatty et al. (2018)).

In order to evaluate whether municipal governments exploited Kalahi to reduce public spending, I construct an original panel data set of quarterly budget allocation and electoral outcomes at the municipal level across the Philippines.³ I estimate intent-to-treat effects among the IPA impact evaluation sample for the years in which grants were given to all treated municipalities (2013 through 2015), retaining all 198 municipalities under their original assignment regardless of compliance status, consistent with the evaluation’s own analytical approach (Beatty et al. (2018)). The outcome for my key specifications is per capita public service spending in Philippine pesos, as impact evaluation data do not allow me to disentangle the specific projects implemented in a given municipality.

On November 8, 2013, midway through program implementation, Typhoon Yolanda (international name Haiyan) struck the Philippines. One of the most powerful tropical cyclones ever recorded, Yolanda killed at least 6,300 Filipinos and is estimated to have cost the Philippine government nearly \$13 billion USD (World Bank (2017)). Given the impact on public spending, I code a dichotomous variable for heavy damage from Yolanda, as determined by the Asian Development Bank (Asian Development Bank (2013)). As the Philippine government continued recovery efforts for years after Yolanda, this variable takes

²For a detailed review of selection criteria for the IPA impact evaluation, see Beatty et al. (2018). More information on project priorities and funds allocation within a given municipality can be found on the website of the Department of Social Welfare and Development (DSWD) of the Philippines, the federal agency that administers Kalahi (DSWD (2024)).

³Data on local government spending are obtained from the Bureau of Local Government Finance of the Philippines. Data on the IPA impact evaluation of Kalahi are publicly available through the Millennium Challenge Corporation Evidence Platform of the ICPSR (IPA (2019)). Grant allocation years and other qualitative information on implementation of the randomized controlled trial are derived from the third round impact evaluation report (Beatty et al. (2018)). Data on municipalities included in the impact evaluation affected by Typhoon Yolanda are obtained from the Kalahi supplementary documentation from the Asian Development Bank (Asian Development Bank (2013)). I merge these data with 2013 municipal election returns published by the Commission on Elections of the Philippines (COMELEC), which are the local election results immediately preceding the intervention.

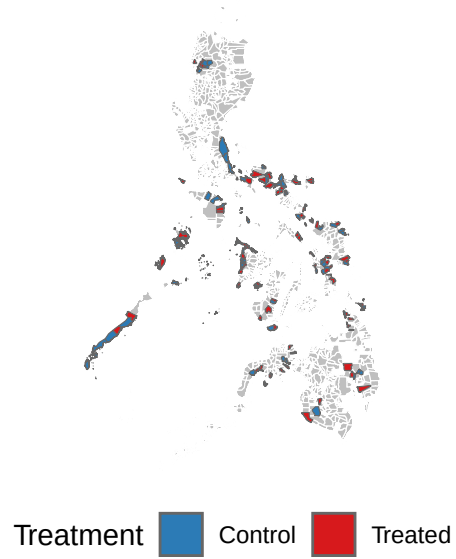
the value of 1 for high-impact municipalities for all quarters in my sample following the storm.

A further consideration bearing on the analysis window concerns the KC-National CDD Program (KC-NCDDP), a nationwide scale-up of Kalahi formally launched in June 2014. By the time of the endline survey in 2015, 79 of 99 control municipalities had been enrolled in KC-NCDDP (Beatty et al. (2018)). Because control municipalities receiving a program functionally identical to the treatment have their own fiscal incentive to substitute external provision for municipal spending, KC-NCDDP enrollment attenuates the treatment-control spending difference from mid-2014 onward, biasing crowd-out estimates toward zero. Spending results using the full period should therefore be interpreted as conservative lower bounds on the true crowd-out effect. To confirm this, I re-estimate all primary specifications restricting the sample to 2013 through the first half of 2014, prior to KC-NCDDP's first disbursement in July 2014, and present results in the appendix. Estimates are stable or larger in the pre-contamination window, consistent with conservative bias in the full-period results.

Treated and control groups are balanced across pre-treatment covariates and spending outcomes. I test balance on all pre-treatment covariates systematically via randomization inference. Results from randomization inference are provided in appendix Table A1.⁴

⁴Randomization inference was conducted by simulating 1,000 random assignments consistent with the procedure employed in the impact evaluation to compute randomization-based p-values for overall and individual covariate balance. Neither the overall p-value nor the p-values of any individual covariates are statistically significant at conventional levels ($p < .1$).

Figure 1: Kalahi Evaluation Sample, Philippines



4 Results

I estimate the causal effect of Kalahi on public spending outcomes with the following equation:

$$y_{it} = \tau D_i + X_{ipt} + \epsilon_{it}$$

where y_{it} represents per capita spending in municipality i at quarter t and D_i is an indicator for Kalahi treatment. The coefficient τ captures the intent-to-treat effect on municipal spending, while X_{ipt} denotes a vector of unit-specific controls, including matched pair and time fixed effects. Standard errors are clustered at the municipality level, the unit of treatment assignment. The remaining term accounts for error.

Table 1 shows that Kalahi significantly reduced municipal public service spending in the same domains targeted by the program. The estimated effect is about PhP 89 per resident per quarter. For a municipality at the sample median population, this corresponds to roughly PhP 9.4 million (\$230,00) less public service spending per year.⁵ This is a substantively large fiscal response: rather than simply adding externally financed projects on top of existing municipal provision, Kalahi induced local governments to withdraw a meaningful amount of their own spending from overlapping service sectors.

The estimated effect on total municipal spending is also negative, suggesting that the decline in program-relevant spending was not fully offset by observable increases elsewhere in the budget. This pattern is consistent with fiscal substitution rather than simple relabeling across standard expenditure categories. Subsequent results using discretionary public procurement data examine whether freed fiscal space was politically redeployed.

A potential concern in program evaluation is that control municipalities bordering treated areas may indirectly benefit from spillovers, such as through interjurisdictional learning, shared infrastructure, or political signaling effects. To account for this, I re-estimate the baseline model including an indicator for control municipalities which border treated municipalities. This specification identifies the direct treatment effect while adjusting for potential indirect exposure to the program.

The results indicate that municipalities that receive the grant spend significantly less from their own budgets in program-relevant categories, even after adjusting for spillovers. Control municipalities which border treatment municipalities show slightly higher spending levels, suggesting modest positive spillovers, possibly due to complementary investments or political imitation effects by neighboring local governments. The mild evidence of positive spillovers in adjacent

⁵The median municipal population in the Kalahi impact evaluation sample is about 26,396. I convert Philippine pesos to dollars using the exchange rate at the beginning of the year in which Kalahi was implemented (2013), which was approximately 40.6 PhP per dollar.

Table 1: Non-governmental development programs crowd-out public spending

	Kalahi spending	Total spending	IRA	Peace Corps	Spillovers
Kalahi	−89.034*** (26.422)	−73.047* (34.007)	−65 500.509 (210 255.464)		−61.349* (26.217)
Peace Corps				14.603 (28.050)	
Num.Obs.	2367	2367	2367	2367	2367
R2	0.695	0.616	0.888	0.689	0.694
R2 Adj.	0.679	0.596	0.882	0.673	0.678
Std.Errors	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X	X	X	X
FE: quarter	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Spending outcomes are measured in per capita Philippine pesos (PhP), except IRA, which is measured in levels. All models include matched pair and quarter fixed effects with standard errors clustered at the municipality level.

municipalities further reinforces that the main effect reflects genuine fiscal substitution by treated localities rather than regional budget displacement.

As a placebo test, I examine the Internal Revenue Allotment (IRA), the constitutionally mandated intergovernmental transfer that finances roughly 80 percent of municipal operating income. The IRA is determined by a fixed formula based on population, land area, and income class. Because IRA allocations are computed centrally and municipalities exercise no discretion over them, treatment should have no effect on IRA receipts. I estimate this specification in levels with log population as a covariate rather than per capita, since the population denominator differs from the census benchmark used in the IRA formula and would otherwise induce ratio bias. The estimated treatment effect is small and statistically insignificant, consistent with the placebo prediction.

As a mechanism test, I estimate the same specification using Peace Corps Philippines volunteer assignments as the treatment variable. Peace Corps volunteers operate in overlapping service domains but disburse no funds, providing no fiscal slack for mayors to exploit. The absence of any significant effect on public service spending is consistent with the fiscal channel. It is the resources Kalahi frees, not the mere presence of an external program, that drives crowd-out.

The preceding results reveal an average crowd-out effect, but theory predicts that the magnitude of this effect should vary across political environments. Specifically, if politicians in competitive districts face stronger incentives to maintain visible public service provision, then crowd-out should be less pronounced where electoral competition is high. Here I present evidence that political competition determines how politicians respond to development assistance.

I measure political competition using the incumbent mayor’s margin of victory in 2013, immediately preceding the main period of municipal fiscal response, with narrower margins interpreted as indicating more intense electoral competi-

tion. I reorient the measure so that higher values correspond to greater competition, then mean-center and standardize it, allowing interaction coefficients to be interpreted in standard-deviation units. Figure 2 illustrates this moderating relationship descriptively by plotting public service spending against political competition by treatment assignment, with fitted LOESS regressions overlaid.

To test this rigorously, I estimate interaction models that include the treatment indicator and municipal political competition. Results are presented in Table 2. The coefficient on the interaction term is positive and statistically significant, indicating that Kalahi reduced public spending primarily in low-competition municipalities. The spending decline is attenuated and may reverse at higher levels of competition, consistent with the theoretical expectation that electoral pressure constrains fiscal substitution.

These results are robust to alternative measures of competition. In the appendix, I re-estimate the main interaction using several alternative constructions of electoral competition: the standardized measure used in the baseline, a mean-centered continuous version, a within-sample rank transformation, and binary indicators for whether the mayoral race was “close” using 10 and 20 percentage-point victory-margin thresholds. Across these specifications, the treatment-by-competition interaction remains substantively similar, indicating that the main result is not driven by any single coding choice for electoral competition.

Given that political competition is not randomly assigned, I investigate additional plausible explanations for these findings to probe that results do not derive from political dynamics correlated with competition. In particular, I include a dummy indicator for political dynasties, a salient feature of Philippine politics known to hamper competition. I also include controls for long-run political structure, including political competition from the 2010 and 2007 municipal elections. Results are presented in the full model and the treatment–competition interaction remains large and precisely estimated. Historical political structure does not condition the fiscal response to treatment, indicating that crowd-out is consistent with contemporaneous electoral incentives rather than persistent political dominance.

To further investigate the mechanisms through which political competition hampers crowd-out, I explore the fiscal behavior of term-limited mayors who lack reelection incentives. In the Philippines, mayors are constitutionally limited to three consecutive terms. Term-limited mayors in 2013, having already served two prior terms, could not seek reelection and thus faced weaker electoral accountability. If political competition disciplines incumbents primarily through reelection concerns, then the moderating effect of competition should disappear when incumbents are term-limited.

To test this, I estimate the baseline model on subsamples of term-limited and reelection-eligible mayors. The results support the hypothesized mechanism. Among incumbents eligible for reelection, competition strongly constrains fiscal substitution. However, this effect disappears when incumbents cannot run again.

Table 2: Competition dampens crowd-out

	Baseline	Full controls	Term-limited	Re-election eligible
treat	−80.657*** (22.860)	−68.178** (21.614)	−30.039 (37.361)	−113.103** (35.611)
treat x competition	199.096** (68.411)	167.527** (59.506)	7.689 (70.107)	322.665** (105.223)
Num.Obs.	2367	2307	552	1815
R2	0.716	0.724	0.889	0.691
R2 Adj.	0.701	0.709	0.877	0.672
Std.Errors	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X	X	X
FE: quarter	X	X	X	X

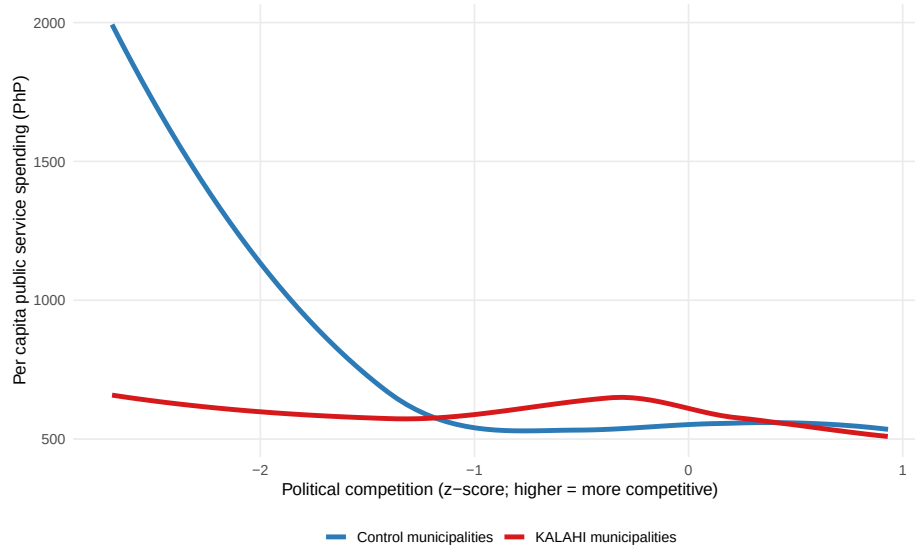
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Spending measured per capita in Philippine pesos. Models include matched pair and quarter-level fixed effects with standard errors clustered at the municipality level.

This pattern suggests that reelection incentives are the channel through which political competition disciplines fiscal behavior. When incumbents' electoral futures are no longer at stake, competitive pressure ceases to moderate the fiscal response to aid.

Figure 2: Smoothed spending curves by treatment status

The treated-control spending difference narrows as political competition increases



5 Freed resources finance political patronage

A large literature in political economy argues that politicians in weakly accountable environments do not simply consume fiscal slack, they invest it strategically in distributive networks that sustain political support (Stokes, Dunning, and Nazareno (2013); Golden and Picci (2008)). In the Philippine context, the canonical form of this investment is clientelism: the targeted distribution of private or club goods to political supporters in exchange for electoral loyalty (Hicken (2011); Hutchcroft (2012); McCoy (2009)). In the Philippines, where broker networks and village-level electoral organization are central to political competition, the geographic distribution of local public spending is a primary instrument of this strategy (Hutchcroft (2012); Cruz, Labonne, and Querubín (2017)). Classic models of aid fungibility treat substitution as the end of the story: external financing relaxes budget constraints, and governments quietly reallocate the savings (Oates (1972); Chatterjee, Giuliano, and Kaya (2012)). But if freed resources flow politically rather than programmatically, the normative implications are unambiguous. Aid would actively subsidize the clientelist strategies of incumbents in low-accountability environments. The efficiency loss from crowd-out is then not a regrettable fiscal byproduct but a predictable political equilibrium.

Procurement is a particularly useful domain for observing political redeployment because it converts municipal resources into geographically traceable contracts. Unlike aggregate budget categories, procurement records identify where public spending is implemented and who receives the immediate benefits of local government projects. Philippine law requires local governments to publish procurement opportunities and award information (“Republic Act 9184” (n.d.)), and the resulting records include project classification, contract value, and the village of implementation. This allows me to examine not only whether Kalahi changed the level of municipal spending, but whether it altered the spatial distribution of discretionary public resources. I exploit this geographic granularity to test whether Kalahi caused municipalities to shift procurement toward villages that were politically supportive of the incumbent.

I code outcomes that capture both the scale and the composition of municipal procurement directed toward politically favorable villages. I aggregate procurement contracts to the municipality-quarter level and construct two measures using village-level electoral returns from the 2013 mayoral election. The first is per capita procurement spending awarded to villages in which the incumbent mayor’s support was above the municipal median. The second is the share of total procurement value in the municipality-quarter awarded to those above-median support villages. I use this within-municipality threshold rather than raw vote share so the estimates capture political targeting relative to the mayor’s own local support distribution, net of differences across municipalities in the incumbent’s overall popularity. Together, these outcomes distinguish whether treated mayors sent more procurement resources to supportive villages in absolute terms and whether they tilted the composition of procurement toward

those villages.

Results are presented in Table 3. Treatment is positive and significant for both outcomes, while the interaction between treatment and competition is negative and significant. These results are robust to alternative support thresholds. This indicates that treated mayors in low-competition municipalities directed more procurement resources toward villages that were relatively supportive of the incumbent, both by increasing per capita procurement spending to those villages and by allocating a larger share of overall procurement value to them. In more competitive municipalities, this pattern is substantially attenuated. The result mirrors the competition gradient documented in the main spending analysis: the same mayors who cut public service spending under Kalahi in low-accountability settings also redirected procurement toward politically favorable constituencies, whereas competitive incumbents allocated procurement less disproportionately. Competition disciplines not only whether crowd-out occurs, but also the political destination of the resources it frees.

Because Kalahi was randomly assigned at the municipality level, treatment effects on municipal-level procurement targeting can be interpreted causally. The remaining inferential challenge is not whether targeting changed, but why. If mayors deploy fiscal space freed by Kalahi through clientelist logic, procurement should shift disproportionately toward villages where the incumbent performed well. However, electorally supportive villages may also differ in need, centrality, economic activity, or ease of service delivery.

I address this in two ways. First, I estimate a contract-level linear probability model with a binary outcome indicating whether contracts were awarded to a village above the median support for the incumbent. I incorporate a battery of pre-treatment village-level covariates, including log population, log land area, population density, distance to the municipal center, detectable nighttime lights in the year prior to Kalahi implementation, and a binary indicator for villages that are the municipal center. These controls help rule out the possibility that contracts flow to incumbent-supporting villages simply because those are larger, denser, more central, more urbanized, more economically active, or easier to serve. Second, I report an IV-style specification that leverages treatment-induced variation in public service spending. The estimate indicates that reductions in public service spending are associated with a larger share of procurement directed toward electorally supportive villages. This evidence is consistent with the causal chain: Kalahi freed fiscal resources that were re-deployed toward political patronage. Results are presented in Appendix Table A5.

I corroborate procurement results with individual-level data from the Philippine Labor Force Survey (LFS), which records employment status, sector, and contract type for a nationally representative sample. Contractual government employment — short-term appointments outside the civil service — is a well-documented channel of political patronage in the Philippines, allowing local executives to reward supporters and mobilize electoral loyalty through public

payrolls that expand and contract with political cycles (McCoy (2009); Robinson and Verdier (2013); Colonnelli, Prem, and Teso (2020)). If mayors are deploying freed resources efficiently in response to genuine community demand, there is no reason to expect systematic increases in casual public sector hiring. If they are converting fiscal slack into political capital, short-term government employment should rise in treated municipalities.

To test this, I estimate a linear probability model predicting the probability that a respondent reports short-term government employment, controlling for respondent demographics and core municipal characteristics employed in the main specifications, with province fixed effects and standard errors clustered at the municipal level. I include province-level fixed effects rather than matched pair as the Labor Force Survey samples only a subset of the municipalities in the Kalahi impact evaluation and provides limited within-pair treatment-control overlap. As shown in Table 3, treated municipalities exhibit a statistically distinguishable pattern of casual government employment, which is also disciplined by political competition.

Taken together, the procurement and employment results triangulate on the same mechanism from independent data sources. Treated municipalities systematically directed public contracts toward electorally supportive villages in low-competition settings and simultaneously expanded casual government employment. These two canonical instruments of political clientelism are difficult to rationalize as efficient public provision. Critically, the procurement targeting and labor force patterns are moderated by competition in precisely the same direction as the main crowd-out result, suggesting that democratic accountability shapes not only the level of public spending but the distributional logic of how freed resources are deployed. This finding connects the fungibility literature to a growing body of work on how democratic institutions alter the composition, not just the level, of public spending (Persson and Tabellini (2002); Keefer (2007); Golden and Picci (2008)). External aid distorts both the quantity and distribution of public resources, and that distortion is politically structured.

These results carry a broader implication that cuts against a common assumption in the development literature. Aid programs, especially community-driven development programs such as Kalahi, are frequently justified not only on welfare grounds but as investments in state capacity: interventions that, by demonstrating the value of public goods provision, might strengthen the programmatic relationship between citizens and the state (Mansuri and Rao (2013); Casey, Glennerster, and Miguel (2012)). The results here suggest the opposite in precisely the settings where capacity-building is most needed. In low-competition municipalities, where mayors are entrenched, accountability is weak, and programmatic politics is already precarious (Keefer (2007); Hicken (2011)), Kalahi appears not to have complemented the state. Instead, it afforded incumbents resources that can deepen those weaknesses. Freed from the need to provide services in Kalahi-covered domains, and insulated from electoral punishment for failing to do so, these mayors expanded patronage payrolls and directed public

contracts toward political supporters. Each of these responses makes the political environment harder to contest: patronage employment binds brokers and voters to the incumbent (Robinson and Verdier (2013); Stokes, Dunning, and Nazareno (2013)), targeted contracts build localized loyalty (Hutchcroft (2012); Cruz, Labonne, and Querubín (2017)), and the withdrawal of programmatic provision may reduce citizens' expectations of the state as a neutral service provider (Keefer (2007); Hicken (2011)).

The concern, then, is that aid may produce not a stronger local state but a more entrenched one, in which the substitution of external assistance for public goods compounds the institutional weaknesses that aid was intended to address. This is not a design failure of Kalahi specifically. It is a structural risk of delivering development assistance through political environments where accountability is weak, and it suggests that the governance context of aid recipients deserves as much attention as the technical design of the programs themselves.

Table 3: Freed funds are targeted to political supporters

	Pesos per capita	Procurement share	Casual gov. employment
treat	1285.525** (410.309)	8.341** (2.881)	0.002* (0.001)
treat x competition	-1418.011** (462.245)	-9.624** (3.318)	-0.002*** (0.000)
Num.Obs.	145	145	74 342
R2	0.563	0.538	0.013
R2 Adj.	0.251	0.208	0.012
Std.Errors	by: mun_psgc	by: mun_psgc	by: mun
FE: quarter	X	X	X
FE: pairnum	X	X	
FE: province			X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

6 Welfare consequences of crowd-out

While fungibility itself presents a concern for development policymakers who wish for aid to finance its intended programs, it does not necessarily harm human development. Governments may allocate freed funds efficiently in order to address pressing domestic needs and optimize resources (Feyzioglu, Swaroop, and Zhu (1996)). I exploit variation in the capacity for mayors to reallocate resources to explore whether this capacity attenuated welfare outcomes from Kalahi.

I develop a municipal fiscal discretion index that captures the extent to which local governments had baseline room to reallocate resources. The index is designed to measure ex ante fiscal capacity for reallocation rather than realized behavior under Kalahi. For the fiscal components, I average all available pre-treatment municipal budget data before 2012, which reduces sensitivity to unusual single-year budget shocks and captures more stable differences in local fiscal structure. The first dimension is fiscal slack: own-source revenue capacity and cash position, which proxy the extent to which municipal governments had resources available to adjust spending without immediate external constraint. Municipalities with stronger local revenue bases and more liquid balances should have faced fewer practical barriers to shifting funds across uses once Kalahi reduced pressure to finance service provision directly.

The second dimension captures how flexibly the municipal budget was already organized. I measure this using a payroll-and-debt component based on pre-treatment government employment and debt burden. The debt measure comes from the same pre-2012 municipal budget histories, while government employment is measured using pre-treatment Labor Force Survey data from 2009-2011. These variables proxy the extent to which local fiscal management was already structured around discretionary or politically manipulable commitments rather than rigid service provision. I standardize each component and average them

into a single municipal fiscal discretion index, signed so that higher values indicate greater baseline scope for discretionary reallocation. The core idea is that crowd-out should be most consequential where local governments had not only the political incentive to substitute away from programmatic provision, but also the fiscal capacity to redeploy the resulting slack.

I focus on the endline results for two reasons. First, it is the more policy-relevant horizon because it captures welfare after implementation had more time to unfold. Second, by endline most control municipalities had been enrolled in the national scale-up, so treatment assignment no longer corresponds cleanly to actual exposure and demands a different estimand. Following Beatty et al., who led the impact evaluation, I instrument realized treatment intensity with original random assignment (Beatty et al. (2018)). Specifically, I use assignment to instrument the share of completed Kalahi cycles observed by endline, as well as its interaction with mayoral discretion. This recovers the effect of exogenous variation in actual program dosage under two-sided noncompliance created by scale-up, rather than conflating original assignment with contaminated control exposure.

I retain matched-pair fixed effects and cluster standard errors at the municipality level, but unlike Beatty et al., I also include pre-treatment municipal covariates. This adjustment is appropriate here because the central estimand is not simply the average program effect, but whether treatment gains were attenuated in municipalities where mayors had greater ex ante capacity to reallocate freed resources. Conditioning on baseline municipal characteristics helps isolate that moderation pattern without altering the assignment-based source of identifying variation in treatment intensity.

I construct a pooled welfare index from the impact evaluation survey to capture the domains most directly affected by Kalahi: service access and economic welfare. I orient all component variables so that higher values correspond to better welfare or broader access, standardize each component relative to the control group, and average them into a single index. This pooled measure reduces noise from individual survey items and avoids placing weight on any one outcome, while preserving a direct connection to the mechanisms through which crowd-out should attenuate program returns. An appendix table lists the component outcomes included in the index.

Formally, I estimate:

$$\begin{aligned} welfare_{ip} &= \beta_1 \widehat{Cycles}_i + \beta_2 (\widehat{Cycles}_i \times Discretion_i) + X_i' \gamma + \alpha_p + \epsilon_{ip} \\ Cycles_i &= \pi_1 Treat_i + \pi_2 (Treat_i \times Discretion_i) + X_i' \delta + \alpha_p + \nu_{ip} \\ Cycles_i \times Discretion_i &= \rho_1 Treat_i + \rho_2 (Treat_i \times Discretion_i) + X_i' \lambda + \alpha_p + \eta_{ip} \end{aligned}$$

where $welfare_{ip}$ denotes a household-level welfare outcome in municipality i and matched pair p , $Cycles_i$ is the realized share of completed Kalahi cycles

by endline, and $Discretion_i$ is the pre-treatment municipal discretion index. Together, the first stage equations instrument realized dosage and its interaction with discretion using original treatment assignment and its interaction with discretion. α_p denotes matched-pair fixed effects, and X_i includes baseline municipal covariates: population, poverty incidence, land area, number of villages, mayoral vote margin, and dynasty status. Standard errors are clustered at the municipality level. The coefficient of interest is β_2 , which captures whether the welfare returns to realized Kalahi exposure were weaker where mayors had greater ex ante scope to reallocate freed resources.

Results, presented in Table 4, show that welfare gains at endline were significantly attenuated in municipalities in which mayors had the capacity to reallocate resources. This pattern does not extend to official government assistance, a category encompassing discretionary government transfers for which co-investment should not necessarily decline. If reallocation favored targeted transfers, individual reports of official assistance should remain stable or increase, making this outcome a useful contrast to service access and economic welfare. These findings are consistent with Kalahi enabling fiscally unconstrained mayors to reduce programmatic goods provision rather than broadly depressing all measured outcomes. Welfare results are directionally similar in the midline survey as well, which I present in the appendix.⁶

Appendix Table A8 shows that the welfare moderation result is not driven by municipalities with sparse pre-treatment fiscal histories or by a single aggregation choice in the discretion index. First, I re-estimate the endline welfare model after requiring municipalities to have at least five or ten pre-treatment fiscal years contributing to the index. The estimated moderation effect remains negative and substantively similar, though precision declines under the strictest coverage requirement. I then hold the main inclusion rule fixed and vary the aggregate index construction, combining the underlying fiscal, debt, and payroll measures in alternative ways. Across these constructions, greater mayoral discretion consistently attenuates Kalahi’s welfare gains.

Substantively, these results suggest that Kalahi’s overall positive effect across a range of welfare outcomes may have been partially offset by fiscal responses from local governments. The findings highlight the importance of considering the political context in evaluating development programs. Even well-designed, community-driven programs may yield muted welfare returns when they inadvertently reshape public resources.

⁶Midline results are estimated with a reduced form regression rather than IV. KC-NCDDP contamination had not yet occurred, so original assignment still corresponds to actual exposure.

Table 4: Endline welfare gains contract where mayors have greater discretion

	Pooled welfare	Official assistance
treatment intensity x mayoral discretion	−0.233** (0.084)	0.015 (0.037)
Num.Obs.	5184	5184
R2	0.454	0.057
R2 Adj.	0.443	0.038
Std.Errors	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: IV estimates with matched-pair fixed effects, municipal covariates, and standard errors clustered at the municipality level.

7 Conclusion

Development programs are often designed to bypass weak governments and avoid leakages. This paper shows that bypassing public budgets does not bypass politics. When non-governmental programs provide goods that are observationally similar to public services, they relax municipal fiscal constraints while making government retrenchment harder for citizens to observe. The result is a political channel of aid fungibility: external provision can free public resources that incumbents redeploy for political advantage.

Leveraging a randomized evaluation of the World Bank–financed Kalahi program in the Philippines, I show that non-governmental development assistance reduced local public spending in the same domains targeted by the program. This crowd-out was concentrated in low-competition municipalities, where incumbents faced weaker electoral accountability; in competitive municipalities, mayors maintained public spending. I then trace the political destination of the resources Kalahi freed. In low-competition municipalities, treated mayors directed procurement contracts toward electorally supportive villages and expanded casual government employment, two canonical instruments of clientelism. By contrast, political competition attenuated these distributional distortions. Finally, survey evidence suggests that these fiscal responses had welfare consequences: Kalahi’s welfare gains were weaker where mayors had greater *ex ante* discretion to reallocate municipal resources.

However, these findings reflect the setting in which Kalahi was implemented. Kalahi operated in relatively poor municipalities in the Philippines, where local governments face binding fiscal constraints, mayors exercise meaningful discretion over spending and procurement, and clientelist networks remain central to political competition. These features make Kalahi especially relevant for observing the mechanism theorized here: external provision relaxes local budget constraints, while weak accountability permits incumbents to redirect the resulting fiscal space. The results should therefore generalize most readily to aid programs that overlap with government service mandates in low- and middle-income democracies where local officials control substitutable resources and electoral accountability is uneven. They should travel less directly to settings where public budgets are rigid, party competition is programmatic, or voters can clearly distinguish donor provision from government performance. The broader implication is not that non-governmental aid necessarily corrodes governance, but that its political effects depend on the accountability institutions and fiscal discretion of the governments whose mandates it overlaps.

These results carry clear implications for development policy. Bypassing government channels may insulate aid from direct capture, but it does not eliminate the political incentives of incumbents to exploit the fiscal space that external provision creates. Monitoring donor outputs alone is therefore insufficient; donors must also track whether governments maintain provision in overlapping service domains. Aid programs operating in low-accountability environments should an-

ticipate substitution effects and design monitoring mechanisms that hold local governments accountable for maintaining co-investment in program-relevant domains. More broadly, the governance context of aid recipients deserves as much analytical attention as the technical design of the programs themselves. Where democratic institutions are weak, the effectiveness of development assistance depends not only on program quality but on whether the political environment permits its gains to survive.

References

- Asian Development Bank. 2013. “SUPPLEMENTARY DOCUMENT 6: TY-PHOON YOLANDA-AFFECTED AREAS AND AREAS COVERED BY THE KALAHİ- CIDSS NATIONAL COMMUNITY-DRIVEN DEVELOPMENT PROJECT.”
- Beatty, Amanda, Ariel BenYishay, Elisabeth King, Aniceto Orbeta, and Menno Pradhan. 2018. “Kalahi-CIDSS Impact Evaluation Third Round Report.” <https://www.elisabethking.com/s/Beatty-et-al-Final-Report-2018.pdf>.
- Bräutigam, Deborah A., and Stephen Knack. 2004. “Foreign Aid, Institutions, and Governance in Sub-Saharan Africa.” *Economic Development and Cultural Change* 52 (2): 255–85. <https://doi.org/10.1086/380592>.
- Burgess, Robin, Remi Jedwab, Edward Miguel, Ameet Morjaria, and Gerard Padró I Miquel. 2015. “The Value of Democracy: Evidence from Road Building in Kenya,” June. <https://doi.org/10.1257/aer.20131031>.
- Casey, Katherine, Rachel Glennerster, and Edward Miguel. 2012. “Reshaping Institutions: Evidence on Aid Impacts Using a Preanalysis Plan*.” *The Quarterly Journal of Economics* 127 (4): 1755–1812. <https://doi.org/10.1093/qje/qje027>.
- Chatterjee, Santanu, Paola Giuliano, and Ilker Kaya. 2012. “Where Has All the Money Gone? Foreign Aid and the Composition of Government Spending.” *The B.E. Journal of Macroeconomics* 12 (1). <https://doi.org/10.1515/1935-1690.2458>.
- Colonnelli, Emanuele, Mounu Prem, and Edoardo Teso. 2020. “Patronage and Selection in Public Sector Organizations.” *American Economic Review* 110 (10): 3071–99. <https://doi.org/10.1257/aer.20181491>.
- Cruz, Cesi, Julien Labonne, and Pablo Querubín. 2017. “Politician Family Networks and Electoral Outcomes: Evidence from the Philippines.” *American Economic Review* 107 (10): 3006–37. <https://doi.org/10.1257/aer.20150343>.
- De Luca, Giacomo, Roland Hodler, Paul A. Raschky, and Michele Valsecchi. 2018. “Ethnic Favoritism: An Axiom of Politics?” *Journal of Development Economics* 132 (May): 115–29. <https://doi.org/10.1016/j.jdeveco.2017.12.006>.
- DSWD. 2024. “Kalahi-CIDSS FAQs.” <https://kalahi.dswd.gov.ph/about/community-driven-development>.
- Ferraz, Claudio, and Frederico Finan. 2011. “Electoral Accountability and Corruption: Evidence from the Audits of Local Governments.” *American Economic Review* 101 (4): 12741311. <https://www.aeaweb.org/articles?id=10.1257/aer.101.4.1274>.
- Feyzioglu, Tarhan, Vinaya Swaroop, and Min Zhu. 1996. *Foreign Aid’s Impact on Public Spending*. World Bank Publications.
- . 1998. “A Panel Data Analysis of the Fungibility of Foreign Aid.” *The World Bank Economic Review* 12 (1): 2958. <https://academic.oup.com/wber/article-abstract/12/1/29/1644642>.
- Franck, Raphaël, and Ilia Rainer. 2012. “Does the Leader’s Ethnicity Matter? Ethnic Favoritism, Education, and Health in Sub-Saharan Africa.” *Amer-*

- ican Political Science Review* 106 (2): 294–325. <https://doi.org/10.1017/S0003055412000172>.
- Golden, Miriam A., and Lucio Picci. 2008. “Pork-Barrel Politics in Postwar Italy, 1953–94.” *American Journal of Political Science* 52 (2): 268–89. <https://doi.org/10.1111/j.1540-5907.2007.00312.x>.
- Harding, Robin, and David Stasavage. 2014. “What Democracy Does (and Doesn’t Do) for Basic Services: School Fees, School Inputs, and African Elections.” *The Journal of Politics* 76 (1): 229–45. <https://doi.org/10.1017/S0022381613001254>.
- Hicken, Allen. 2011. “Clientelism.” *Annual Review of Political Science* 14 (Volume 14, 2011): 289–310. <https://doi.org/10.1146/annurev.polisci.031908.220508>.
- Hines, James R., and Richard H. Thaler. 1995. “The Flypaper Effect.” *Journal of Economic Perspectives* 9 (4): 217–26. <https://doi.org/10.1257/jep.9.4.217>.
- Hutchcroft, Paul. 2012. “Re-Slicing the Pie of Patronage: The Politics of the Internal Revenue Allotment in the Philippines, 1991–2010.” https://openresearch-repository.anu.edu.au/bitstream/1885/28935/2/01_Hutchcroft_Re-Slicing_the_pie_of_2012.pdf.
- IPA. 2019. “Millennium Challenge Corporation Evidence Platform,” April. <https://doi.org/10.3886/sn1t-fv11>.
- Keefer, Philip. 2007. “Clientelism, Credibility, and the Policy Choices of Young Democracies.” *American Journal of Political Science* 51 (4): 804–21. <https://doi.org/10.1111/j.1540-5907.2007.00282.x>.
- Lindbeck, Assar, and Jörgen W. Weibull. 1987. “Balanced-Budget Redistribution as the Outcome of Political Competition.” *Public Choice* 52 (3): 273–97. <https://doi.org/10.1007/BF00116710>.
- Lizzeri, Alessandro, and Nicola Persico. 2001. “The Provision of Public Goods Under Alternative Electoral Incentives.” *American Economic Review* 91 (1): 225–39. <https://doi.org/10.1257/aer.91.1.225>.
- Mansuri, Ghazala, and Vijayendra Rao. 2013. *Localizing Development: Does Participation Work?* World Bank Publications.
- McCoy, Alfred W. 2009. *An Anarchy of Families: State and Family in the Philippines*. Univ of Wisconsin Press.
- Moss, Todd J., Gunilla Pettersson Geland, and Nicolas van de Walle. 2006. “An Aid-Institutions Paradox? A Review Essay on Aid Dependency and State Building in Sub-Saharan Africa,” January. <https://doi.org/10.2139/ssrn.860826>.
- Oates, Wallace E. 1972. “Fiscal Federalism.” *HarcourtBrace Jovanovich Inc.*
- Olken, Benjamin A., and Rohini Pande. 2012. “Corruption in Developing Countries.” *Annual Review of Economics* 4 (Volume 4, 2012): 479–509. <https://doi.org/10.1146/annurev-economics-080511-110917>.
- Persson, Torsten, and Guido Tabellini. 2002. *Political Economics: Explaining Economic Policy*. MIT Press.
- “Republic Act 9184.” n.d. https://lawphil.net/statutes/repacts/ra2003/ra_9184_2003.html.

- Robinson, James A., and Thierry Verdier. 2013. "The Political Economy of Clientelism." *The Scandinavian Journal of Economics* 115 (2): 260–91. <https://doi.org/10.1111/sjoe.12010>.
- Shleifer, Andrei, and Robert W. Vishny. 1993. "Corruption*." *The Quarterly Journal of Economics* 108 (3): 599–617. <https://doi.org/10.2307/2118402>.
- Solé-Ollé, Albert, and Pilar Sorribas-Navarro. 2008. "The Effects of Partisan Alignment on the Allocation of Intergovernmental Transfers. Differences-in-Differences Estimates for Spain." *Journal of Public Economics*, New directions in fiscal federalism, 92 (12): 2302–19. <https://doi.org/10.1016/j.jpubeco.2007.06.014>.
- Stokes, Susan C. 2009. "Political Clientelism." In, edited by Carles Boix and Susan C. Stokes, 0. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780199566020.003.0025>.
- Stokes, Susan C., Thad Dunning, and Marcelo Nazareno. 2013. *Brokers, Voters, and Clientelism*. Cambridge University Press.
- World Bank. 2017. "Philippines - Lessons Learned from Typhoon Yolanda : An Assessment of the Post-Yolanda Short and Medium-Term Recovery and Rehabilitation Interventions of the Government." <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/537771507750178892/Philippines-Lessons-learned-from-typhoon-Yolanda-an-assessment-of-the-post-Yolanda-short-and-medium-term-recovery-and-rehabilitation-interventions-of-the-Government>.

Appendix

Table A1: Pretreatment covariate balance

var	mean_treat	mean_cont	sd_treat	sd_cont	p
public service spending	219.3076	258.3637	162.2841	274.6745	0.2585
total spending	1587.4168	1721.8399	1250.7574	1581.4931	0.2876
poverty index	46.5610	46.3275	9.0415	9.5221	0.8630
num barangay	26.1466	24.6040	17.2734	13.1972	0.5993
population	31258.0739	29628.4323	20557.3549	19554.0854	0.7763
land area	192.5472	186.1958	192.5050	195.1503	0.8880

Note: Individual p-values come from robust regressions with matched-pair fixed effects and province-clustered standard errors. Omnibus randomization-inference p-value from 1,000 within-pair reassignments = 0.828.

Table A2: Appendix Table A2: Crowd-out robustness to outcome construction

	Per-capita level	z-score per-capita
Kalahi treatment	−61.349* (26.217)	−0.111* (0.047)
Num.Obs.	2367	2367
R2	0.694	0.694
R2 Adj.	0.678	0.678
Std.Errors	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X
FE: quarter	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Outcome construction varies by column. All models include matched pair and quarter fixed effects with standard errors clustered at the municipality level.

Table A3: Non-governmental development programs crowd-out public spending

	Kalahi spending	Total spending	Spending by competition
treat	−76.218** (26.615)	−82.021* (31.716)	−68.125** (23.726)
treat x competition			195.763** (74.869)
Num.Obs.	1179	1179	1179
R2	0.692	0.687	0.714
R2 Adj.	0.661	0.655	0.685
Std.Errors	by: mun_psgc	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X	X
FE: quarter	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Spending outcomes measured in Philippine pesos (PhP) per capita. Models include matched pair and quarter fixed effects with standard errors clustered at the municipality level.

Table A4: Appendix Table A4: Competition moderator robustness

	z-scored competition	centered competition	rank competition	close election (10pp)	close election (20pp)
Kalahi treatment	−80.657*** (22.860)	−80.657*** (22.860)	−319.696** (116.344)	−135.311** (42.798)	−179.129* (76.436)
Treat x competition (z)	199.096** (68.411)				
Treat x competition (centered)		721.623** (247.957)			
Treat x competition (rank)			482.782* (197.722)		
Treat x close election (10pp)				151.364* (65.599)	
Treat x close election (20pp)					162.563+ (95.104)
Num.Obs.	2367	2367	2367	2367	2367
R2	0.716	0.716	0.702	0.695	0.697
R2 Adj.	0.701	0.701	0.687	0.680	0.681
Std.Errors	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X	X	X	X
FE: quarter	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Each column uses an alternative competition construction in the interaction with treatment. All models include matched pair and quarter fixed effects with standard errors clustered at the municipality level.

Table A5: Appendix Table A5: Procurement targeting robustness across support thresholds

	Contract: above median	IV: above-median share	Top tercile: pesos pc	Top tercile: share	Top quartile: pesos pc	Top quartile: share	Top decile: pesos pc	Top decile: share
trst	6.670* (2.589)		1493.082** (468.216)	6.214* (2.626)	1561.120*** (444.947)	7.199** (2.527)	1517.251*** (389.890)	6.847** (2.344)
treat x competition	-7.714* (2.979)		-1673.039** (533.096)	-7.166* (3.018)	-1734.873** (505.513)	-7.988** (2.888)	-1693.731*** (440.010)	-7.637** (2.676)
freed public service spending		0.001*** (0.000)						
Num.Obs.	497	145	145	145	145	145	145	145
R2	0.316	0.457	0.416	0.441	0.423	0.463	0.417	0.440
R2 Adj.	0.211	0.112	-0.001	0.032	0.012	0.080	0.053	0.055
Std.Errors	by: mun_page	by: mun_page	by: mun_page	by: mun_page	by: mun_page	by: mun_page	by: mun_page	by: mun_page
FE: pairmun	X	X	X	X	X	X	X	X
FE: quarter	X	X	X	X	X	X	X	X

* p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: Column 1 estimates the contract-level above-median support model with barangay-level controls for 2010 population, area, population density, distance to the poblacion or municipal center, an indicator for detectable baseline 2012-2013 nighttime lights, and an indicator for poblacion barangays. Column 2 reports an exploratory IV model of above-median procurement share, instrumenting freed public service spending with Kahabi assignment. Columns 3-8 aggregate procurement to the municipality-quarter level and replicate the main-text interaction specification using alternative top-tercile, top-quartile, and top-decile support thresholds. All models use contracts awarded on or after June 30, 2013 and include pair and quarter fixed effects with standard errors clustered at the municipality level.

Table A6: Appendix Table A6: Pooled welfare moderation across survey waves

	Midline reduced form	Endline IV
treat x mayoral discretion	−0.156** (0.053)	
treatment intensity x mayoral discretion		−0.233** (0.084)
Num.Obs.	2023	5184
R2	0.436	0.454
R2 Adj.	0.422	0.443
Std.Errors	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: The outcome is a row-wise pooled index averaging service access and economic welfare. The midline model is reduced form. The endline model instruments realized treatment intensity and its interaction with mayoral discretion using original assignment and assignment interacted with discretion. Both models include matched-pair fixed effects, municipal covariates, and standard errors clustered at the municipality level.

Table A7: Pooled welfare index component mapping

Wave	Survey outcomes
Midline	Distance to public services; travel time to public services; travel cost to public services; no Yolanda hardship; no other hardship
Endline	Distance to public services; travel time to public services; amount paid for water; time spent fetching water; number of water trips; household spending; labor income

Table A8: Appendix Table A8: Mayoral discretion robustness

	Main index	Main: >=5 fiscal years	Main: >=10 fiscal years	Fiscal slack + debt	Fiscal admin index	All components
treatment intensity x discretion	-0.233** (0.084)	-0.228* (0.094)	-0.259+ (0.144)	-0.223** (0.071)	-0.222** (0.078)	-0.193* (0.083)
Num.Obs.	5184	4795	3628	5184	5184	5184
R2	0.454	0.472	0.529	0.453	0.451	0.449
R2 Adj.	0.443	0.461	0.519	0.442	0.441	0.438
Std.Errors	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc	by: mun_psgc
FE: pairnum	X	X	X	X	X	X

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Note: All models use the endline pooled welfare outcome and mirror the Table 4 IV specification. Columns 1-3 retain the main index construction while varying the required number of fiscal pre-treatment years. Columns 4-6 retain the main data inclusion rule while varying the aggregate index construction.